

Signed Rational Numbers: Subtraction & Sign Errors

Answer Key

Grade 6–7 Mathematics

Quick Answers

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|-----------------|----------------------------|
| 1. -13 | 5. $<$ |
| 2. 5 | 6. 10 |
| 3. -7 | 7. (b) $= -\frac{5}{4}, -$ |
| 4. (b) $= 2, -$ | 8. $-\frac{1}{4}$ |
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What is verified

6 of 8 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problems 4, 7. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 6–7 Mathematics

7.NS.A.1 – problems 1, 2, 3, 4, 5, 6, 7, 8

Difficulty 1–4 of 5 across 10 tagged checks.

Common wrong answers

1. If they got 3: changed the subtraction into addition
 2. If they got -19 : ignored the double negative and subtracted 12
 3. If they got -23 : dropped the opposite sign in front of negative 8
 4. If they got -10 : treated subtracting negative 6 as subtracting positive 6
 6. If they got -8 : subtracted both numbers inside the parentheses instead of distributing the minus
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1. Evaluate: $-5 - 8$

Solution: Rewrite the subtraction as adding the opposite: $-5 - 8 = -5 + (-8)$. Both numbers are negative, so add the absolute values ($5 + 8 = 13$) and keep the negative sign.

-13

2. Evaluate: $-7 - (-12)$

Solution: Subtracting -12 is the same as adding 12: $-7 - (-12) = -7 + 12$. The signs differ, so subtract $12 - 7 = 5$ and keep the sign of the number with the larger absolute value, which is $+12$.

5

3. Evaluate: $-(-8) - 15$

Solution: The opposite of -8 is 8 , so $-(-8) = 8$. Then $8 - 15 = 8 + (-15) = -7$. -7

4. Jorge computes $-4 - (-6)$ and writes $-4 - (-6) = -10$. (a) What error did Jorge make? (b) Find the correct value of $-4 - (-6)$.

Solution: (a) Jorge treated subtracting -6 as subtracting positive 6 : he computed $-4 - 6 = -10$. Subtracting a negative means *adding*, so the double negative should have become $+6$. A correct one-sentence answer must say that $-(-6)$ is $+6$ (or that subtracting a negative is adding).

(b) $-4 - (-6) = -4 + 6 = 2$. 2

5. Write $<$ or $>$ in the blank: $-3 - 5$ $\square$ $-3 - (-5)$

Solution: Left side: $-3 - 5 = -8$. Right side: $-3 - (-5) = -3 + 5 = 2$. Since -8 is to the left of 2 on the number line, the true statement is

$-3 - 5 < -3 - (-5)$

6. Evaluate: $3 - (2 - 9)$

Solution: The minus in front of the parentheses applies to *every* term inside: $3 - (2 - 9) = 3 - 2 + 9$. Left to right: $3 - 2 = 1$, then $1 + 9 = 10$. (Check by working inside first: $2 - 9 = -7$, and $3 - (-7) = 10$.) 10

7. Maya computes $-\frac{1}{2} - \frac{3}{4}$ and writes $-\frac{1}{2} - \frac{3}{4} = \frac{1}{4}$. (a) Explain Maya's error. (b) Find the correct value.

Solution: (a) Maya *added* $\frac{3}{4}$ instead of subtracting it: she computed $-\frac{1}{2} + \frac{3}{4} = \frac{1}{4}$. The expression subtracts a positive $\frac{3}{4}$, so the answer must be *less* than $-\frac{1}{2}$, not positive. A correct explanation names the subtraction-turned-addition error.

(b) With a common denominator of 4 : $-\frac{1}{2} = -\frac{2}{4}$, so $-\frac{2}{4} - \frac{3}{4} = -\frac{5}{4}$. $-\frac{5}{4}$

8. **Challenge.** Evaluate: $-\frac{3}{4} - (-\frac{5}{6}) + (-\frac{1}{3})$

Solution: Rewrite every subtraction as adding the opposite: $-\frac{3}{4} + \frac{5}{6} - \frac{1}{3}$. Use the common denominator 12 :

$$-\frac{9}{12} + \frac{10}{12} - \frac{4}{12} = \frac{-9+10-4}{12} = -\frac{3}{12}.$$

Simplify by dividing top and bottom by 3 , giving

$-\frac{1}{4}$